

PNR In-Depth Physical Design Guide

Best Clean Version from Handwritten Notes + Real-Time EDA Visuals +
Complete Setup-to-Tapeout Flow

Source Topic	Polished Coverage
PNR inputs	.lib/.db, LEF, Tech LEF, ITF/TLU+, SDC, netlist, scan DEF
Routing basics	metal width, spacing, pitch, tracks, vias, layer direction
Implementation	floorplan, power planning, placement, CTS, routing
Signoff	STA, extraction, IR/EM, DRC, LVS, density, ECO, tapeout
Low power	UPF, voltage areas, power switches, isolation, level shifters, tie cells

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28nm / 32nm PnR Training Notes

How This Version Uses Your Source PDF

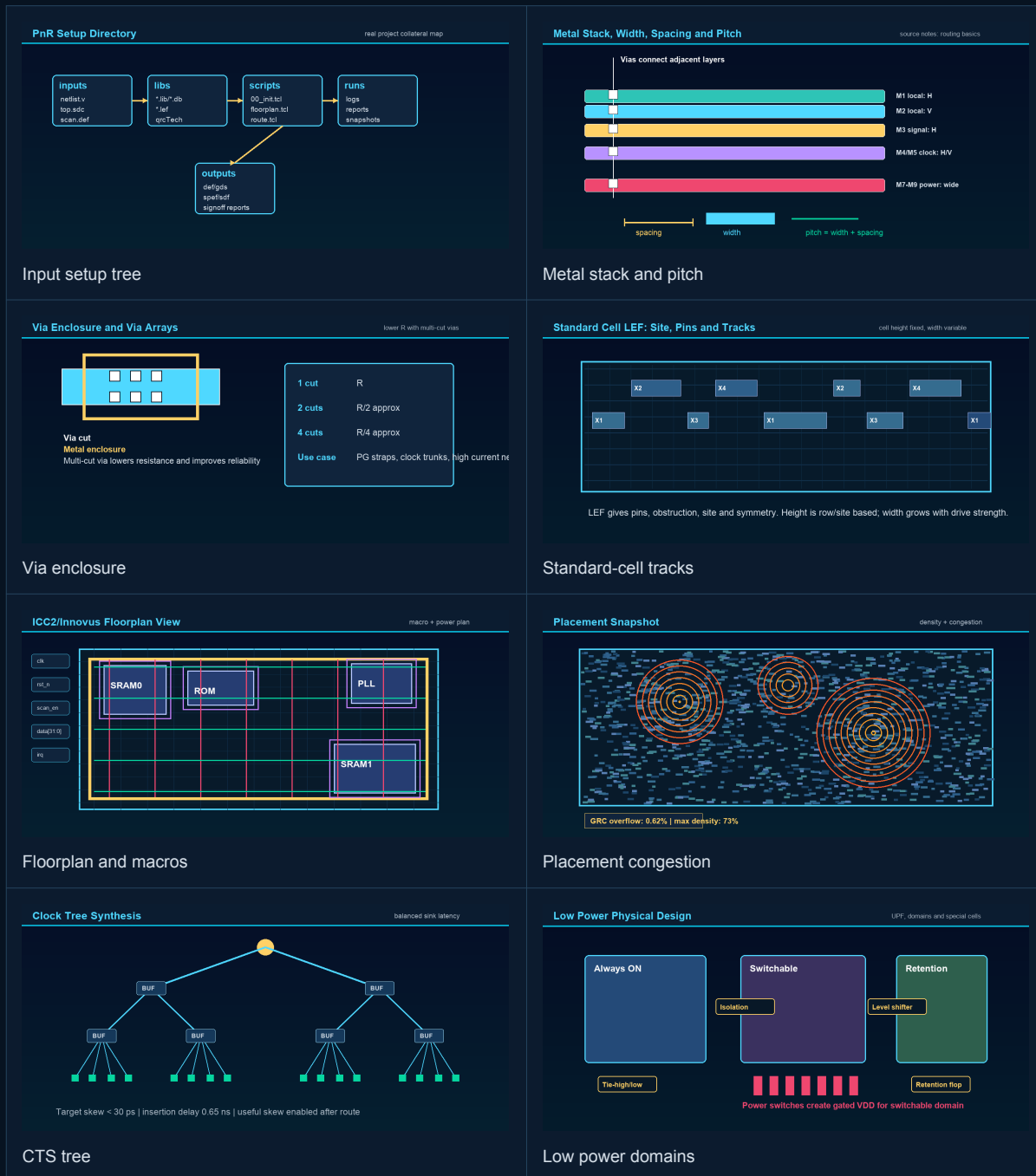
Your source PDF is a 198-page handwritten PnR notebook. Because it is image-based, the content was read visually and with OCR sampling, then converted into a typed, structured guide. The technical sequence and topics come from the source: PnR inputs, metal rules, tracks, vias, macros, floorplan, power planning, placement, timing, CTS, routing, extraction, physical verification, filler/decap/tie cells, ECO and low-power design.

Goal	Create a clean workshop-quality guide that is easier to study than handwritten notes.
Style	Dark EDA-themed pages, real-time visual diagrams, concise tables and commands.
Assumption	28nm/32nm block-level physical design using ICC2/Innovus-style concepts.
Use	Read one section at a time, then practice the command/checklist in a tool lab.

Golden flow: inputs -> MMMC -> floorplan -> power plan -> physical-only cells -> placement -> CTS -> routing -> extraction -> STA -> IR/EM -> DRC/LVS/density -> ECO -> tapeout.

Real-Time Visual Gallery

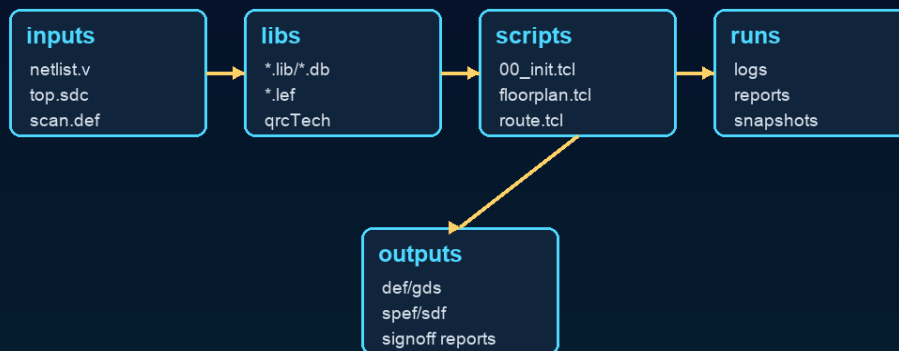
This guide uses clean EDA-style visuals plus source-derived diagrams. They explain the handwritten notes in a way that is easier to revise, print and teach.



1. PnR Inputs: What the Tool Needs

PnR Setup Directory

real project collateral map



Clear Explanation

The source notes begin with the exact PnR inputs: technology-specific data and design-specific data. PnR cannot start only from a netlist; it needs timing, physical, routing, extraction and constraint views that all describe the same cells.

Key Concepts from Source Notes

.lib / .db	Logical/timing library. Contains timing arcs, power, capacitance, setup/hold checks and operating corners.
LEF	Physical library. Contains cell/macro size, pins, obstructions and routing blockages.
Tech LEF	Technology rules: metal layers, preferred direction, width, spacing, pitch, vias and routing grid.
ITF / TLU+ / QRC	Interconnect technology file used for RC extraction and delay calculation.
SDC	Clock, IO delay, uncertainty, false path and multicycle constraints.
Gate netlist	Synthesized Verilog from synthesis.
Scan DEF	Optional DFT placement/scan chain information from synthesis/DFT.

Engineer Checklist

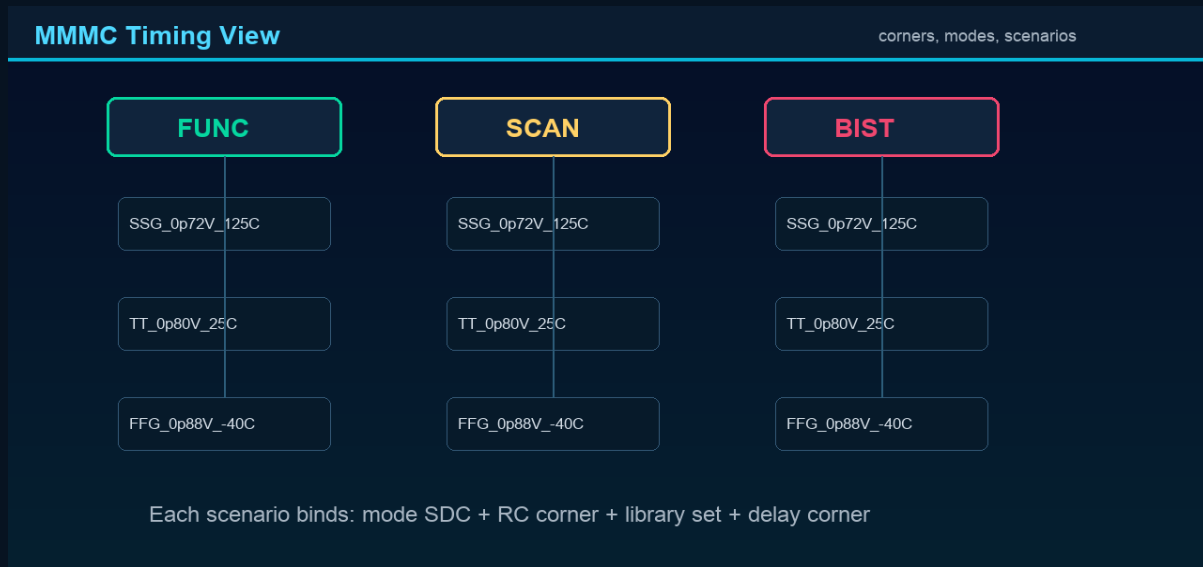
- Check every referenced cell exists in both timing and physical libraries.
- Confirm top module name, clocks and generated clocks.
- Treat warnings about missing pins, black boxes and unconstrained paths as blockers.

Example Lab Commands / Checks

```

read_verilog inputs/top.v
read_lef inputs/tech.lef
read_lef inputs/stdcell.lef
read_liberty inputs/lib/ssg.lib
read_sdc inputs/top.sdc
check_design -summary
  
```

2. PnR Flow Stages



Clear Explanation

The handwritten flow lists floorplanning, placement plus pre-CTS optimization, CTS plus post-CTS optimization, routing plus post-route optimization, and ECO. A real project adds extraction, STA, physical verification and final handoff around these core steps.

Key Concepts from Source Notes

Floorplan	Create die/core, ports, macro placement, power grid and physical-only cells.
Placement	Place standard cells, optimize timing/congestion and legalize rows.
CTS	Build clock tree, balance skew/latency and fix post-CTS timing.
Routing	Connect all nets with legal metals/vias and repair DRC/antenna.
ECO	Engineering change order for timing, DRC, functional or minor design fixes.

Engineer Checklist

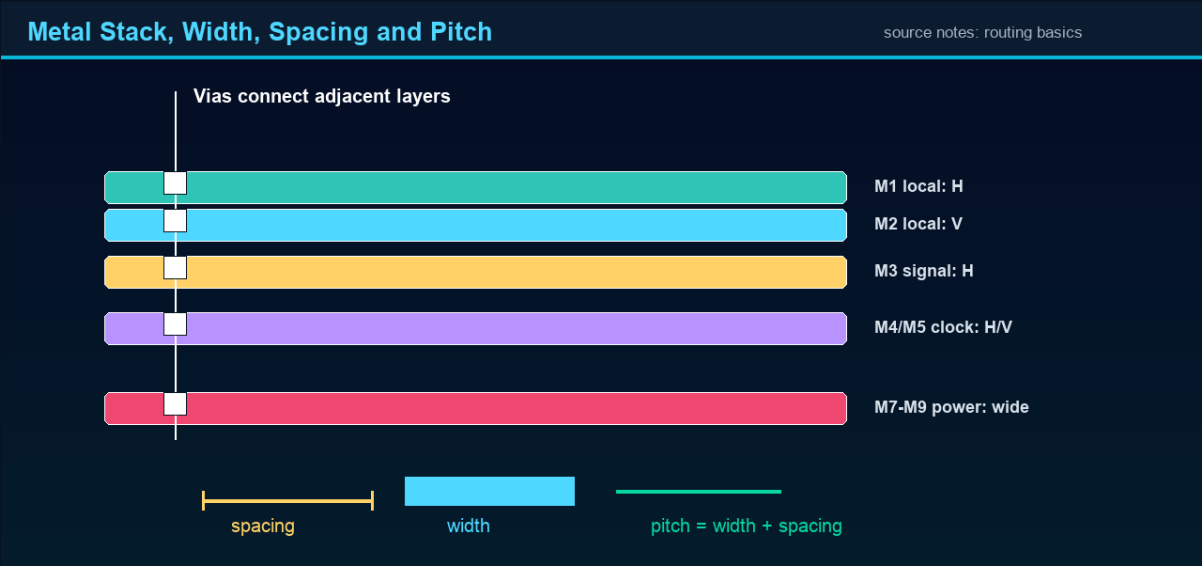
- Each stage must produce reports and a saved database.
- Do not move forward with unresolved structural issues.
- Use incremental ECOs near signoff.

Example Lab Commands / Checks

```

source 01_setup.tcl
source 02_floorplan.tcl
source 03_place_opt.tcl
source 04_cts.tcl
source 05_route_opt.tcl
source 06_signoff.tcl
  
```

3. Routing Terminology: Metal, Width, Spacing, Pitch



Clear Explanation

Routing connects pins using metal shapes. Each metal layer has minimum width, maximum width rules, spacing rules and pitch. Pitch is the center-to-center distance between adjacent tracks, normally equal to width plus spacing.

Key Concepts from Source Notes

Width	Physical width of a wire. Long or high-current nets need wider wires to reduce resistance.
Spacing	Minimum gap between metal shapes. Spacing depends on width, length and layer.
Pitch	Track-to-track distance. Larger pitch gives easier routing but fewer available tracks.
Layer direction	Adjacent layers usually alternate horizontal and vertical preferred directions.

Engineer Checklist

- Use lower layers for dense local routing.
- Use middle layers for clocks and longer signals.
- Use top thick metals for power and long trunks.

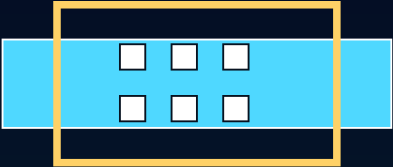
Example Lab Commands / Checks

```
# Conceptual checks
report_routing_rules
report_tracks
check_routes -drc
report_congestion -by_layer
```

4. Via Rules and Via Resistance

Via Enclosure and Via Arrays

lower R with multi-cut vias



Via cut
Metal enclosure
Multi-cut via lowers resistance and improves reliability

1 cut	R
2 cuts	R/2 approx
4 cuts	R/4 approx
Use case	PG straps, clock trunks, high current nets

Clear Explanation

A via connects one metal layer to the next. The notes emphasize via enclosure, cut size and via resistance. For important nets, multiple via cuts are used because parallel vias reduce resistance and improve EM reliability.

Key Concepts from Source Notes

Via cut	The actual contact opening between two metals.
Via enclosure	Required metal overlap around the cut.
Via array	Multiple cuts in parallel; reduces resistance and improves current handling.
Via spacing	Foundry-defined spacing between cuts and nearby metals.

Engineer Checklist

- Use via arrays for PG straps and clock trunks.
- Check via enclosure DRC after route ECOs.
- Do not ignore single-via EM warnings on high-current nets.

Example Lab Commands / Checks

```
set_route_zrt_common_options -post_detail_route_redundant_via_insertion high
insert_zrt_redundant_vias
check_routes > reports/via_route_check.rpt
```

5. Standard Cell LEF, Tracks and Sites

Standard Cell LEF: Site, Pins and Tracks

cell height fixed, width variable

LEF gives pins, obstruction, site and symmetry. Height is row/site based; width grows with drive strength.

Clear Explanation

Standard cell height is fixed by the row/site and track count, while width varies with drive strength. The LEF view describes pins and obstructions. Rows align cells so pins are reachable by routing tracks.

Key Concepts from Source Notes

Site	Legal placement grid for standard cells.
Track cell	Cell height described in routing tracks, such as 7-track or 9-track cells.
Drive strength	Higher drive cells are usually wider.
Pins	Metal pin shapes defined in LEF; pin access strongly affects routability.

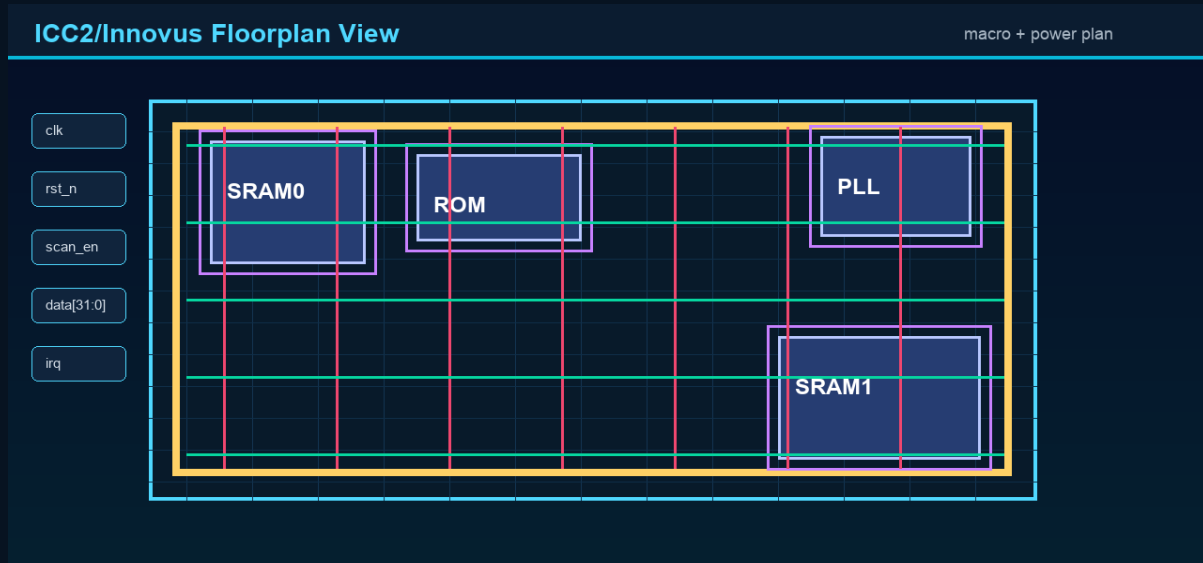
Engineer Checklist

- Check row orientation and site compatibility.
- Watch pin access for high-density cells.
- Do not mix incompatible cell heights in the same row region.

Example Lab Commands / Checks

```
create_placement -floorplan
check_placement -site_row
report_lib_cells -physical
```

6. Macros: Hard, Soft, Halos and Channels



Clear Explanation

The notes distinguish macro concepts: hard macros are fixed pre-designed blocks such as SRAM/ROM/PLL, while soft macros are flexible logical blocks. In PnR, hard macro placement controls congestion, timing and power integrity.

Key Concepts from Source Notes

Hard macro	Fixed layout and pins. You place it but do not edit inside it.
Soft macro	Logical block whose internal placement can still be changed.
Halo	Reserved whitespace around macro to avoid cell crowding.
Channel	Routing space between macros or between macro and boundary.

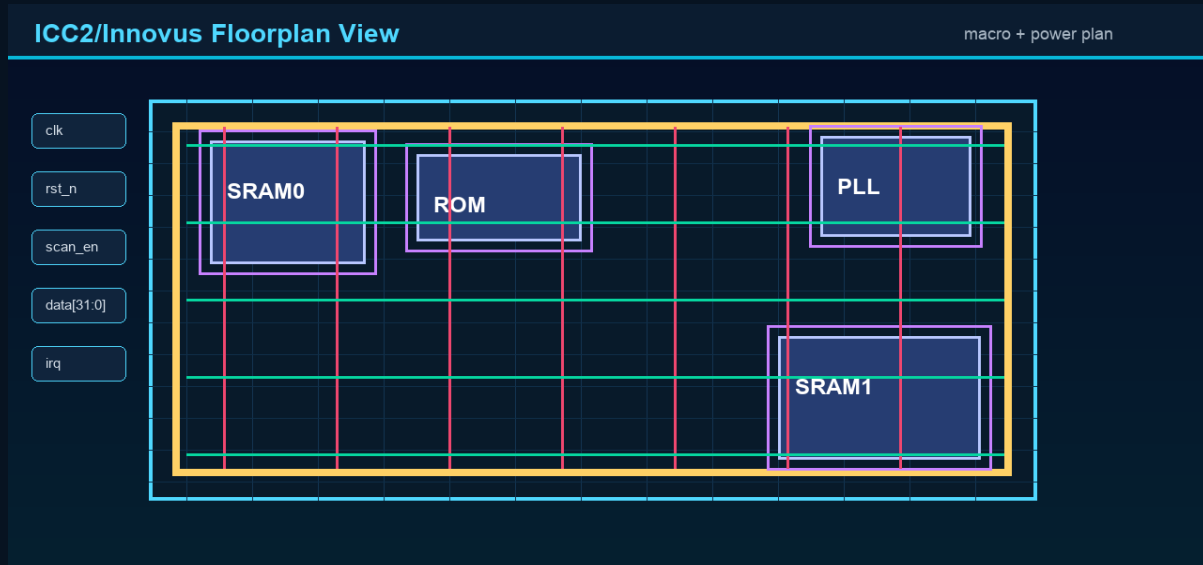
Engineer Checklist

- Place pins toward standard-cell logic.
- Avoid narrow routing channels.
- Add halos and routing blockages before global placement.

Example Lab Commands / Checks

```
place_macro -macro_name SRAM0 -location {200 1500} -orientation R0
create_keepout_margin -outer {10 10 10 10} [get_cells SRAM0]
create_routing_blockage -layers {M1 M2 M3} -boundary $macro_bbox
```

7. Floorplanning: Die, Core, Ports and Utilization



Clear Explanation

Floorplanning creates die area, core area, IO/port placement, macro placement, power grid and pre-placed physical cells. Utilization must leave enough whitespace for optimization, clock buffers and routing.

Key Concepts from Source Notes

Die boundary	Total physical chip/block boundary.
Core boundary	Area where standard cells are placed.
Utilization	Standard-cell area divided by available core area.
Ports	Placed based on connectivity, timing criticality and top-level integration.

Engineer Checklist

- Start around 60-75% utilization for routability.
- Group related ports by interface.
- Run early congestion check before placement signoff.

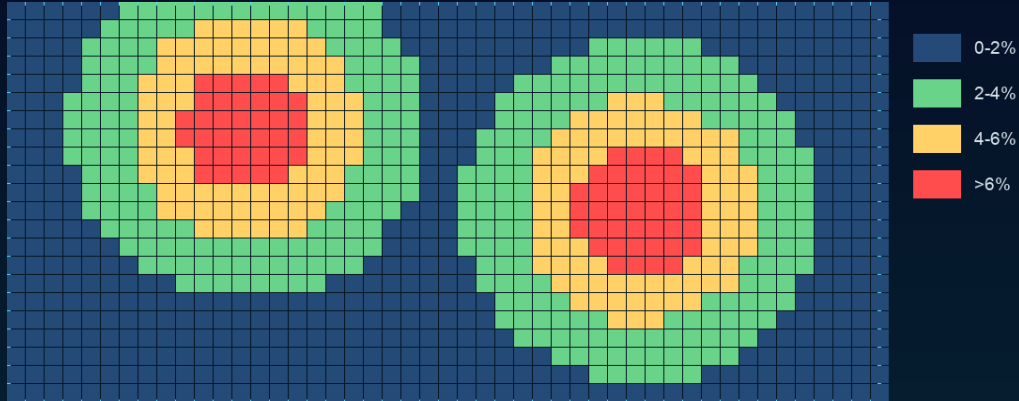
Example Lab Commands / Checks

```
initialize_floorplan -core_utilization 0.70 -aspect_ratio 1.0
place_pins -self
check_floorplan -verbose
report_utilization
```

8. Power Planning: Rings, Straps and Layer Strategy

IR Drop / EM Heatmap

power integrity



Static IR target < 5% VDD. Dynamic peak checked around clock launch and high-toggle windows.

Clear Explanation

The source notes ask why top layers are used for power, middle layers for clock and lower layers for standard-cell routing. Top metals are thicker and lower resistance, so they carry VDD/VSS across the chip with less IR drop. Middle layers are good for controlled clock routing, while lower layers handle dense local signal routing.

Key Concepts from Source Notes

Top layers	Power mesh/rings/straps, high current, low resistance.
Middle layers	Clock trunks and long controlled routes.
Lower layers	Dense local routes near standard cells.
Sroute	Special route connects PG pins, straps, rings and standard-cell rails.

Engineer Checklist

- Connect macro PG pins explicitly.
- Run PG connectivity before placement closure.
- Balance power strap density against signal routability.

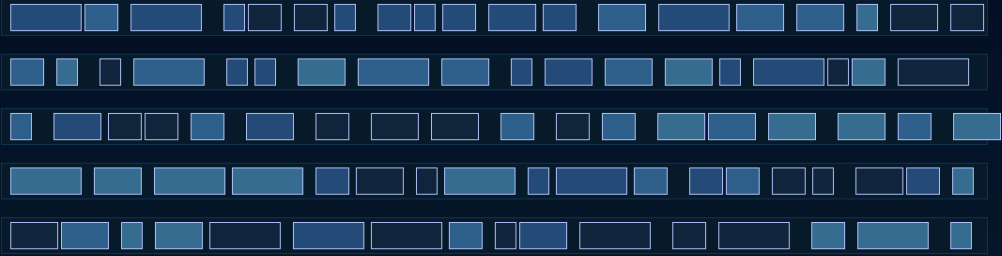
Example Lab Commands / Checks

```
create_pg_ring -nets {VDD VSS} -layers {M8 M9}  
create_pg_stripe -nets {VDD VSS} -layer M7 -pitch 50  
connect_pg_net -automatic  
check_pg_connectivity
```

9. Physical-Only Cells: Tap, Endcap, Filler, Decap, Tie

Fill, Decap, Tie Cells and Density

physical-only signoff cells



Filler: continues well/implant and closes row gaps

Decap: local charge reservoir for dynamic IR

Tie: safe constant 1/0 connection

Dummy fill: meets metal density/CMP rules

Clear Explanation

The later source pages cover filler cells, decap fillers, tie cells and continuity. These cells are not RTL logic, but they are mandatory for manufacturability, power stability and safe constant connections.

Key Concepts from Source Notes

Tap cells	Tie wells/substrate to prevent latch-up.
Endcaps	Protect row edges and macro boundaries.
Filler cells	Maintain well/implant continuity after placement.
Decap cells	Provide local charge reservoir for dynamic IR.
Tie cells	Provide legal constant high/low instead of directly tying gates to rails.

Engineer Checklist

- Insert taps before placement signoff.
- Insert filler after route/ECO.
- Use tie-high/tie-low cells for constants.

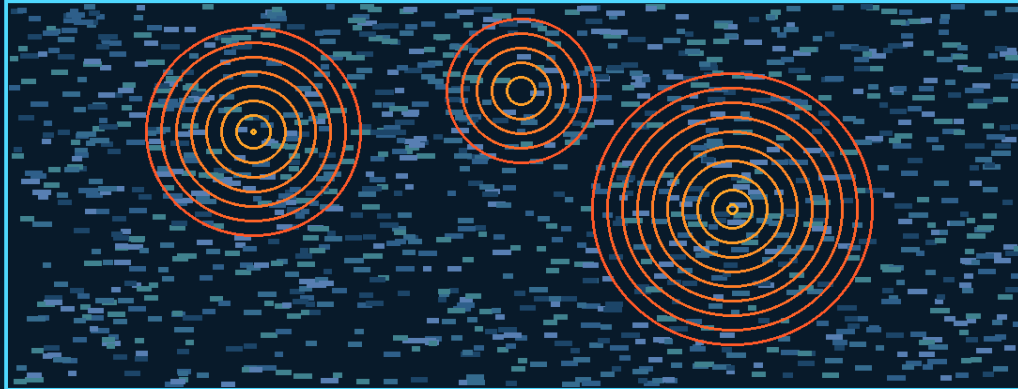
Example Lab Commands / Checks

```
create_tap_cells -distance 25
create_boundary_cells
insert_stdcell_filler -cell_without_metal {FILL1 FILL2}
insert_decap_cells -target_density 5
```

10. Placement and Pre-CTS Optimization

Placement Snapshot

density + congestion

**GRC overflow: 0.62% | max density: 73%**

Clear Explanation

Placement puts all standard cells into rows. Pre-CTS timing uses ideal clocks, so the tool focuses on data paths, fanout, congestion, transition and capacitance. Good placement makes CTS and routing easier.

Key Concepts from Source Notes

Global placement	Coarse cell spreading and wirelength optimization.
Legalization	Cells snapped to legal rows without overlap.
Detailed placement	Local improvement for timing and congestion.
Pre-CTS optimization	Fix data path timing before real clocks are built.

Engineer Checklist

- Check congestion maps.
- Fix high-fanout nets.
- Ensure placement legality before CTS.

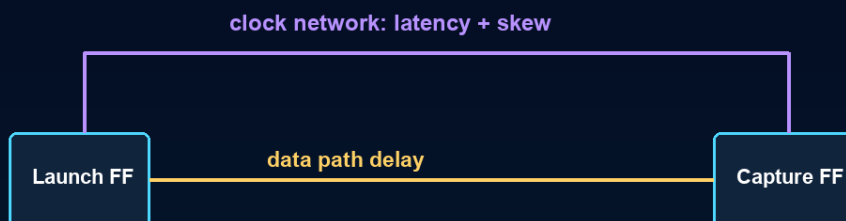
Example Lab Commands / Checks

```
place_opt -effort high
legalize_placement
refine_placement
report_timing > reports/pre_cts_timing.rpt
report_congestion > reports/place_congestion.rpt
```

11. Timing Fundamentals: Setup, Hold, Skew and CRPR

Timing Basics: Setup, Hold, Skew and CRPR

pre-CTS to post-CTS reality



Setup check: launch clk + CK->Q + data delay + setup ≤ capture clk + period
 Hold check: launch clk + CK->Q + min data delay ≥ capture clk + hold
 Useful skew can help setup, but too much skew can hurt hold.
 CRPR removes pessimism from common launch/capture clock path.

Clear Explanation

The source notes around CTS explain skew, useful skew and common clock path pessimism. Timing closure is not just WNS numbers: the engineer must understand launch clock, capture clock, data delay, clock latency and common path pessimism removal.

Key Concepts from Source Notes

Setup	Data must arrive before the capture edge with setup margin.
Hold	Data must not change too soon after the capture edge.
Skew	Difference between launch and capture clock arrival.
Useful skew	Intentional skew that helps setup or hold.
CRPR	Removes pessimism from common clock path delay.

Engineer Checklist

- Check setup and hold separately.
- Useful skew helps one side but can hurt another.
- Enable OCV/AOCV/POCV as required by node.

Example Lab Commands / Checks

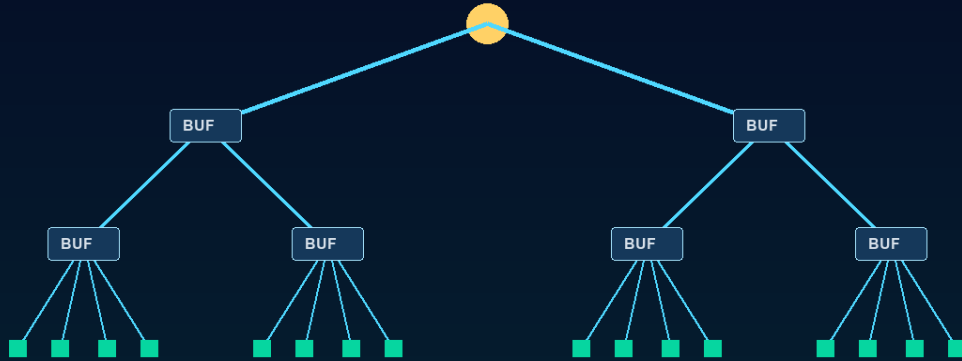
```

set_propagated_clock [all_clocks]
set_timing_derate -early 0.95 -late 1.05
report_timing -delay_type max
report_timing -delay_type min
  
```

12. CTS: Clock Tree Synthesis

Clock Tree Synthesis

balanced sink latency



Target skew < 30 ps | insertion delay 0.65 ns | useful skew enabled after route

Clear Explanation

CTS builds a real clock network. It inserts clock buffers/inverters, balances sink latency, controls skew and transition, and then routes clocks. Source notes emphasize balancing clock paths and using useful skew carefully.

Key Concepts from Source Notes

Goal	Balanced clock latency, low skew and clean transition.
Clock cells	Use clock-qualified buffers/inverters.
NDR/shielding	Wider spacing or shields for critical clocks.
Generated clocks	Must be defined correctly before CTS.

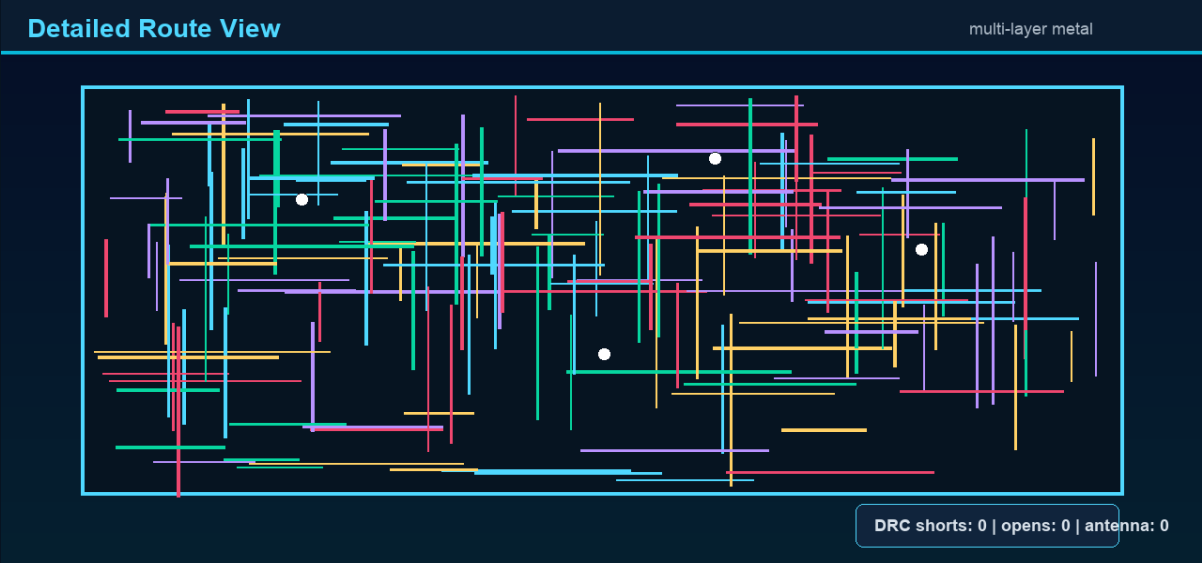
Engineer Checklist

- Check clock-gating timing.
- Check skew and insertion delay.
- Fix post-CTS hold without destroying setup.

Example Lab Commands / Checks

```
set_clock_tree_options -target_skew 0.030
clock_opt -from build_clock -to route_clock
report_clock_tree -summary
report_timing -delay_type min_max
```

13. Routing, Antenna and Post-Route Optimization



Clear Explanation

Routing connects every net with legal wires and vias. Long routes need wider metals or upper layers to reduce resistance and transition. After route, the design is optimized again using routed parasitics.

Key Concepts from Source Notes

Global route	Plans approximate paths and congestion.
Track assignment	Maps routes to legal tracks.
Detailed route	Creates final wire/via shapes.
Antenna	Plasma charging risk during manufacturing.
Post-route opt	Fix timing, SI, electrical and routing issues.

Engineer Checklist

- No real shorts/opens.
- Antenna clean or diode/layer-hop fixed.
- Post-route timing uses extracted parasitics.

Example Lab Commands / Checks

```
route_auto
route_detail
check_routes
route_opt -post_route -setup
route_opt -post_route -hold
```

14. RC Corners, Extraction and STA

Timing Signoff Dashboard

PrimeTime/Tempus style

Scenario	WNS	TNS	Viol	Max Tran	Max Cap
FUNC_SS_125C	+0.018	0.000	0	clean	clean
FUNC_FF_-40C	+0.041	0.000	0	clean	clean
SCAN_SS_125C	+0.006	0.000	0	clean	clean
HOLD_FF_-40C	+0.012	0.000	0	clean	clean
MIN_PULSE_ALL	+0.027	0.000	0	clean	clean

close_timing uses extracted SPEF, propagated clocks, OCV/AOCV, SI delta delay and final SDC.

Clear Explanation

The source notes mention RC corners: typical, worst setup and best hold. Extraction converts actual routed metals and vias into resistance/capacitance. Signoff STA uses these parasitics across MMMC scenarios.

Key Concepts from Source Notes

Cmax / worst RC	Larger capacitance/resistance, setup-sensitive.
Cmin / best RC	Smaller delays, hold-sensitive.
Typical RC	Useful for nominal power/timing analysis.
SPEF	Parasitic file generated from route database.
STA	Checks setup, hold, constraints, pulse width and exceptions.

Engineer Checklist

- No unconstrained endpoints.
- Read correct SPEF per scenario.
- Review false paths and multicycle paths manually.

Example Lab Commands / Checks

```
extract_parasitics -rc_corner Cmax -output top.Cmax.spef
extract_parasitics -rc_corner Cmin -output top.Cmin.spef
read_parasitics top.Cmax.spef
report_global_timing
```

15. DRC, LVS, Density and Metal Fill

Calibre/PVS Physical Verification

DRC/LVS closure



Initial run: 384 DRC | Final signoff: 0 DRC / 0 LVS

Clear Explanation

The source pages near the end discuss density and dummy metal fill. DRC checks manufacturing rules, LVS checks layout versus netlist, density checks CMP/manufacturing requirements and metal fill is added to satisfy density rules.

Key Concepts from Source Notes

DRC	Width, spacing, enclosure, via, density and pattern rules.
LVS	Layout connectivity equals schematic/netlist connectivity.
Density	Minimum/maximum metal area in foundry windows.
Metal fill	Dummy metal added to meet density; affects parasitics.

Engineer Checklist

- Run DRC/LVS after final ECO.
- Run density/fill checks.
- Re-extract and re-run timing after fill.

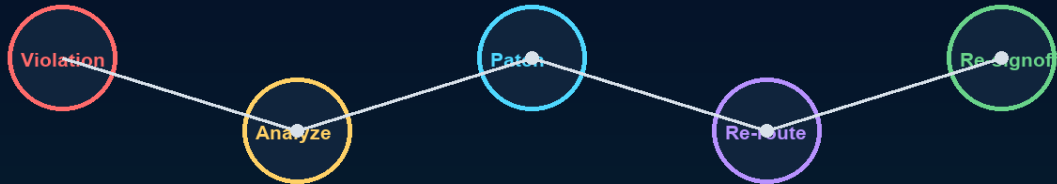
Example Lab Commands / Checks

```
stream_out outputs/top.gds
calibre -drc signoff/drc.rule
calibre -lvs signoff/lvs.rule
insert_metal_fill -timing_driven true
extract_parasitics -real_metalfill_extraction true
```

16. ECO: Functional, Timing and DRC Fixes

ECO Loop

timing to clean GDS



Metal-only ECO preferred after route. Re-run timing, DRC, LVS, antenna, extraction and IR delta.

Clear Explanation

The source notes define ECO as engineering change order and split it into functional ECO and timing ECO. Near tapeout, ECOs must be small, traceable and re-signed because every change can create new timing or physical verification issues.

Key Concepts from Source Notes

Functional ECO	Logic change from design/verification requirement.
Timing ECO	Buffer/resize/delay fixes for setup, hold, transition or cap.
DRC ECO	Route or geometry fix for physical violation.
Metal-only ECO	Late fix using spare cells and upper metal changes.

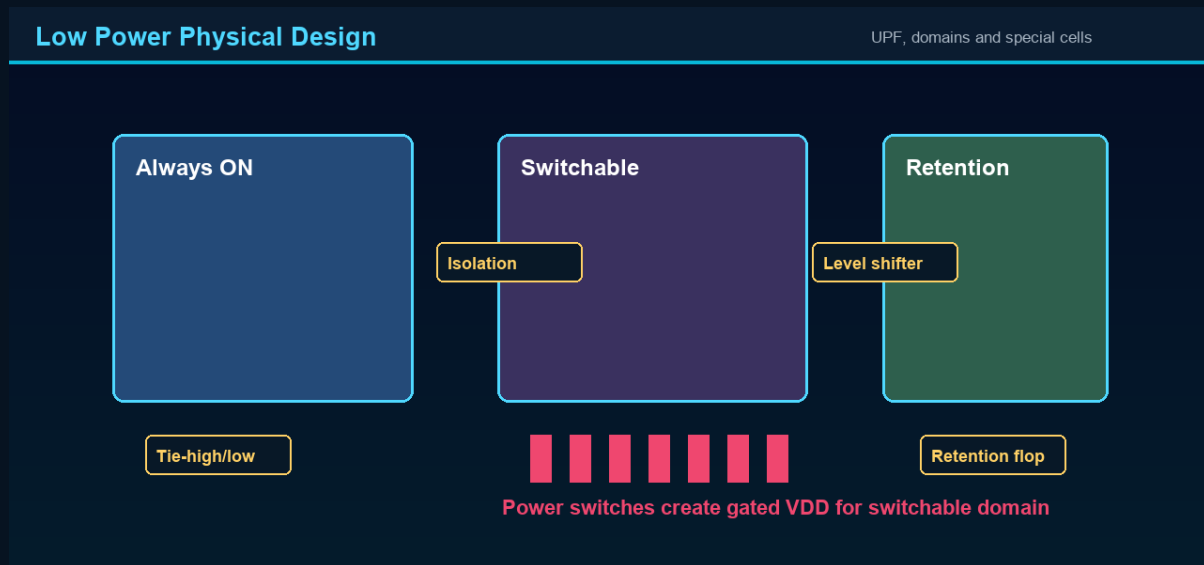
Engineer Checklist

- Script every ECO.
- Compare pre/post timing and DRC.
- Re-run affected signoff checks.

Example Lab Commands / Checks

```
eco_netlist -compare old.v new.v
place_eco_cells
route_eco
report_timing > reports/eco_timing.rpt
check_routes > reports/eco_route.rpt
```

17. Low Power PnR: UPF, Domains and Power Switches



Clear Explanation

The final source pages cover power switches, voltage domains and special cells. Low-power physical design uses UPF to define domains, always-on logic, isolation, level shifters, retention cells and switchable supplies.

Key Concepts from Source Notes

UPF	Power intent: domains, supplies, isolation, retention and level shifting.
Power switch	Header/footer device that creates gated VDD/VSS.
Always-on cell	Logic powered even when another domain is off.
Isolation cell	Clamps outputs from powered-down domains.
Level shifter	Safely crosses voltage domains.
Retention flop	Stores state during power-down.

Engineer Checklist

- Create voltage areas before placement.
- Place switches and always-on routes carefully.
- Check isolation/level shifter insertion and PG connectivity.

Example Lab Commands / Checks

```
read_upf inputs/top.upf
create_voltage_area PD_SW -coordinate {...}
insert_power_switches
connect_pg_net -automatic
check_mv_design
```

18. Final Tapeout Checklist

Timing Signoff Dashboard

PrimeTime/Tempus style

Scenario	WNS	TNS	Viol	Max Tran	Max Cap
FUNC_SS_125C	+0.018	0.000	0	clean	clean
FUNC_FF_-40C	+0.041	0.000	0	clean	clean
SCAN_SS_125C	+0.006	0.000	0	clean	clean
HOLD_FF_-40C	+0.012	0.000	0	clean	clean
MIN_PULSE_ALL	+0.027	0.000	0	clean	clean

close_timing uses extracted SPEF, propagated clocks, OCV/AOCV, SI delta delay and final SDC.

Clear Explanation

The best PnR flow ends with a controlled release package. The final database, GDS/OASIS, netlist, DEF, SPEF, SDF, SDC, reports, waivers and checksums must all correspond to the same frozen design state.

Key Concepts from Source Notes

Timing	Setup/hold clean across all scenarios.
Power	IR/EM clean or waiver-clean.
Physical	DRC/LVS/antenna/density clean.
Database	Final DEF, Verilog, GDS/OASIS, SPEF/SDF saved.
Release	Checksums, waiver log and signoff summary included.

Engineer Checklist

- Freeze final database.
- Generate checksums.
- Archive logs/reports with run tag.
- Get PD, STA, PV, DFT and project owner approval.

Example Lab Commands / Checks

```
write_def outputs/top_final.def
write_verilog outputs/top_final.v
write_gds outputs/top_final.gds
write_sdf outputs/top_final.sdf
md5sum outputs/* > outputs/release_checksums.md5
```